

Development Boards That Are Making IoT DESIGNING Simpler

DEEPSHIKHA SHUKLA

The IoT is developing at a rapid pace, and so is the need for small, inexpensive computing hardware. An IoT board must incorporate advanced functionalities in a tiny package

The Internet of Things (IoT) has been described as the fourth industrial revolution by design engineers. It has changed the way we live, by way of smart and self-driving cars, smart cities, drones, intelligent robots, smart grids, the IoT for industrial and agriculture applications, and so much more.

IoT boards incorporate advanced functionalities into a tiny size, which is shrinking forever. In this article we discuss what an IoT board designer needs to focus on for best results.

Optimising spaces

Every inch of layout space needs to be optimised for all tracks, components and vias. It is about integrating maximum functionality into one board, such as compact size, enhanced data transfer speed, display, sensors, power supply and more, to achieve functionality and flexibility. Every year, we are entering a new design domain that includes high-density interconnect

and rigid-flex PCBs. We are living in a market scenario where just good enough PCBs will no longer be accepted. Meeting the needs of this IoT era of flexible electronics requires working with new materials for designs that we can wear on our bodies or even ingest.

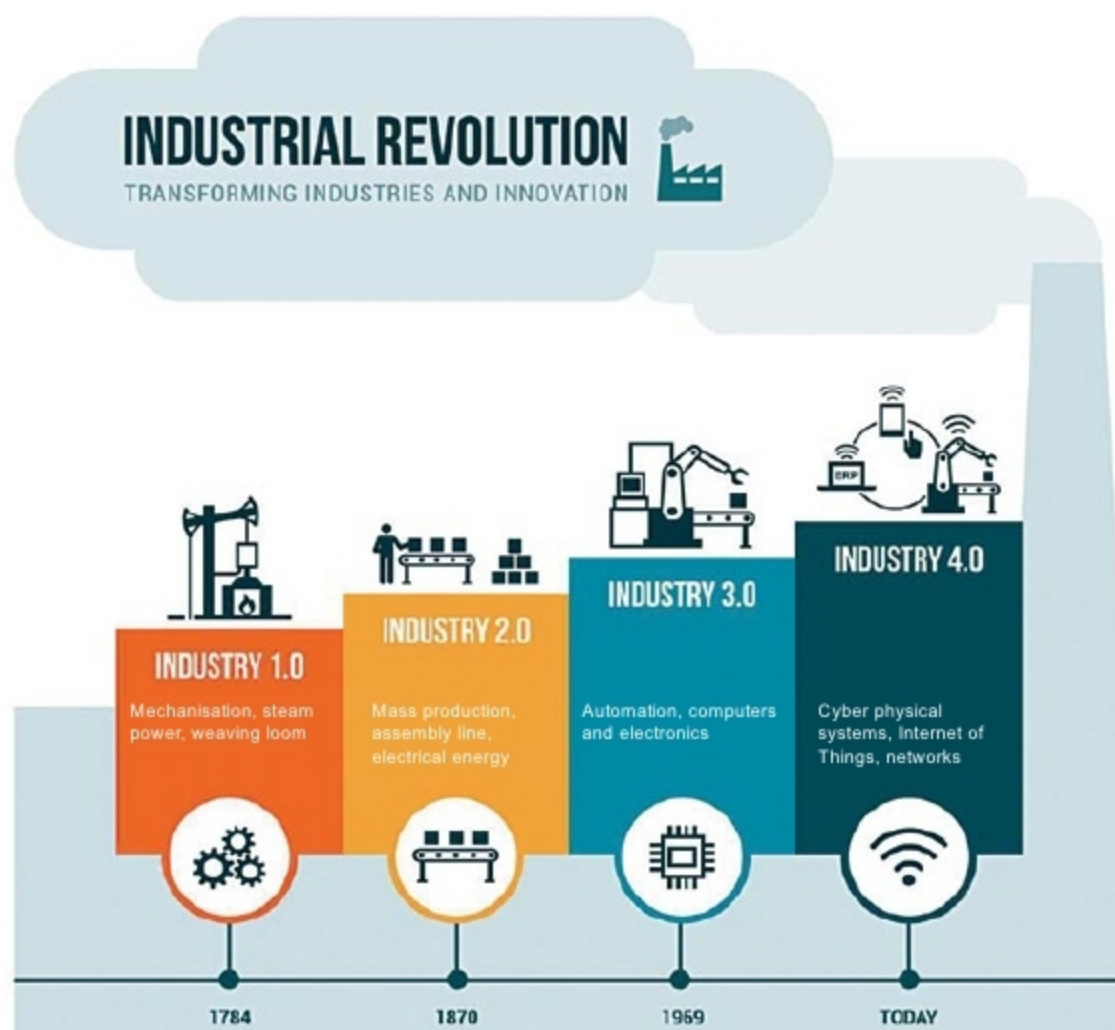
Sunil Deshmukh, senior general manager, technical operation, Millennium Semiconductors, says, "We design smart IoT boards for all segments, including smart homes such as smart lighting, smart switches, wireless controls for home appliances, security systems, energy-saving systems; industrial IoT (IIoT) such as machine control and power-saving systems, data acquisition and management, IoT gateways, production control, industrial safety and RF communications. These boards include sensors, signal conditioning features, smart logics, peripheral controls, wireless communications (Wi-Fi and GSM) and power supplies."

He adds, "Recently-launched Internet-enabled smart air-coolers can be operated with an app—can control 10 air-coolers at a time. Similar features and functionalities can be applied to other devices or appliances. However, there are challenges with the ecosystem for the development of the IoT. Pricing of product is a challenge. Components have to be imported from China or Taiwan. This creates dependency and limitation for designs."

New technologies

To keep things slim and thin, designers and manufacturers are collaboratively working on new packaging technologies. They are making the combination of surface-mount and through-hole components technologies redundant. To make integration of digital logic, analogue and RF systems into a single chip seamless, designers are increasingly using system-in-package technique.

Multi-chip module technique helps keep the form factor thin, by connecting multiple ICs on a single die. Three-dimensional IC packaging allows stacking of multiple silicon dies in a single device. This gives a smaller footprint and reduces power consumption.



Industrial revolution (Credit: www.nordigi.no)